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# The Zep Re Academy Parametric Insurance Webinar – 05/06/2024

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# Day 1 Training Objectives

1. Discuss the state of agriculture insurance globally and in Africa
2. Discuss the insurability of agriculture risks
3. Describe the various risk classifications of agriculture insurance
4. Apply the risk classifications of agriculture insurance in product design

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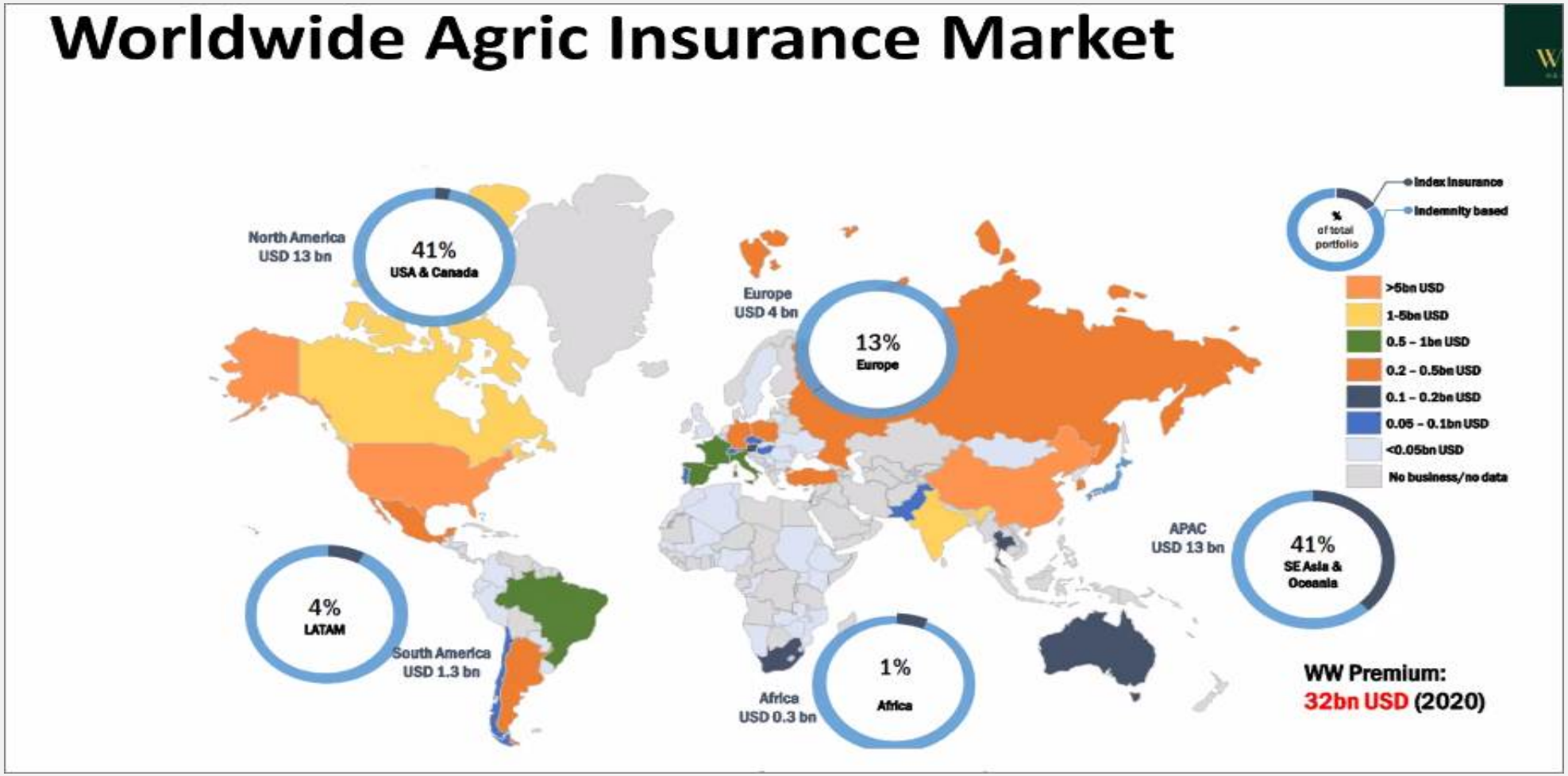
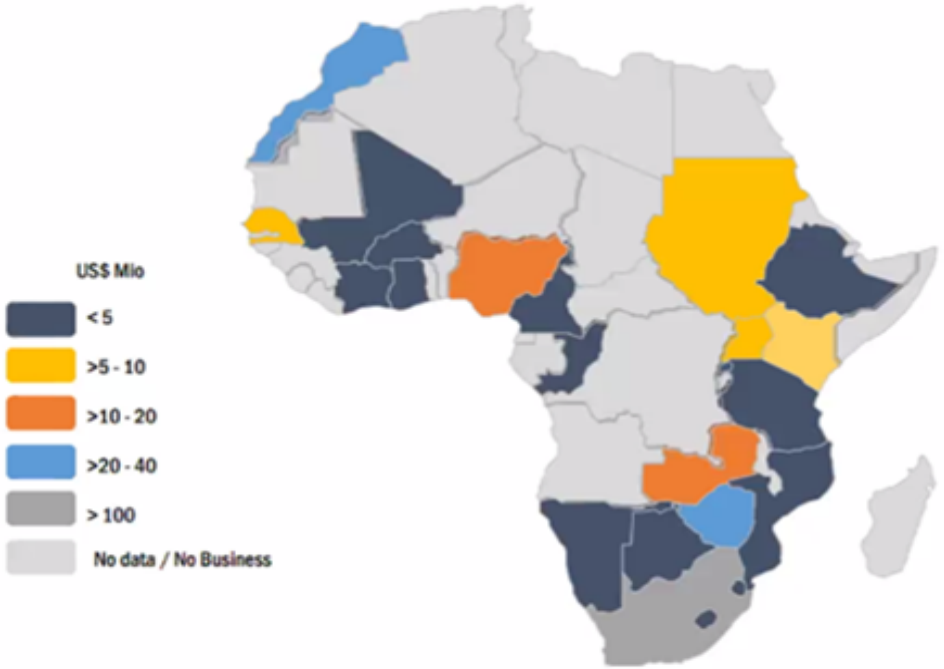
# Global Agric. Insurance premiums



# Africa Agriculture GWP

## Africa Agriculture Premium Production




**Current situation**

- Agricultural Insurance premium on the continent approx. 300mn USD in 2020
- Southern Africa has the longest experience starting early 20th Century
- Southern and Eastern Africa contribute more than 80% of premium
- Eastern Africa has a good recent development trajectory

**Protection Gap to close**

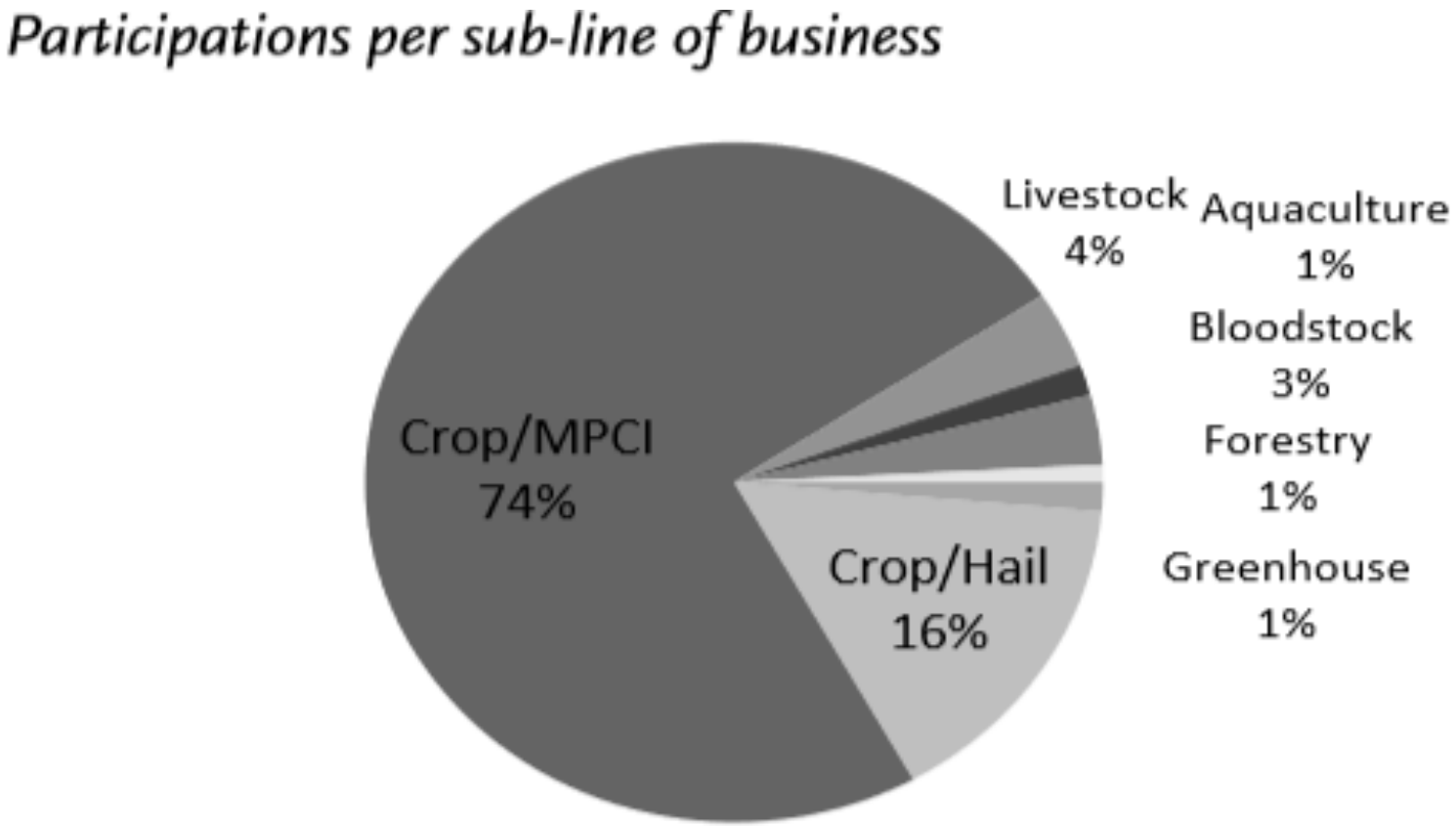
Significant improvements necessary:

- Product suitability
- Affordability
- Value chain inefficiencies
- Distribution
- Information and Training

**Potential**

- Potential of 1bn USD within 10 years, provided that penetration can be substantially increased

# Distribution by product globally




Source: World Bank

- Crop insurance is the most developed line accounting for about 90% of the written premium worldwide.
- MPCI dominant in the USA and Canada, Named peril dominant in Europe.

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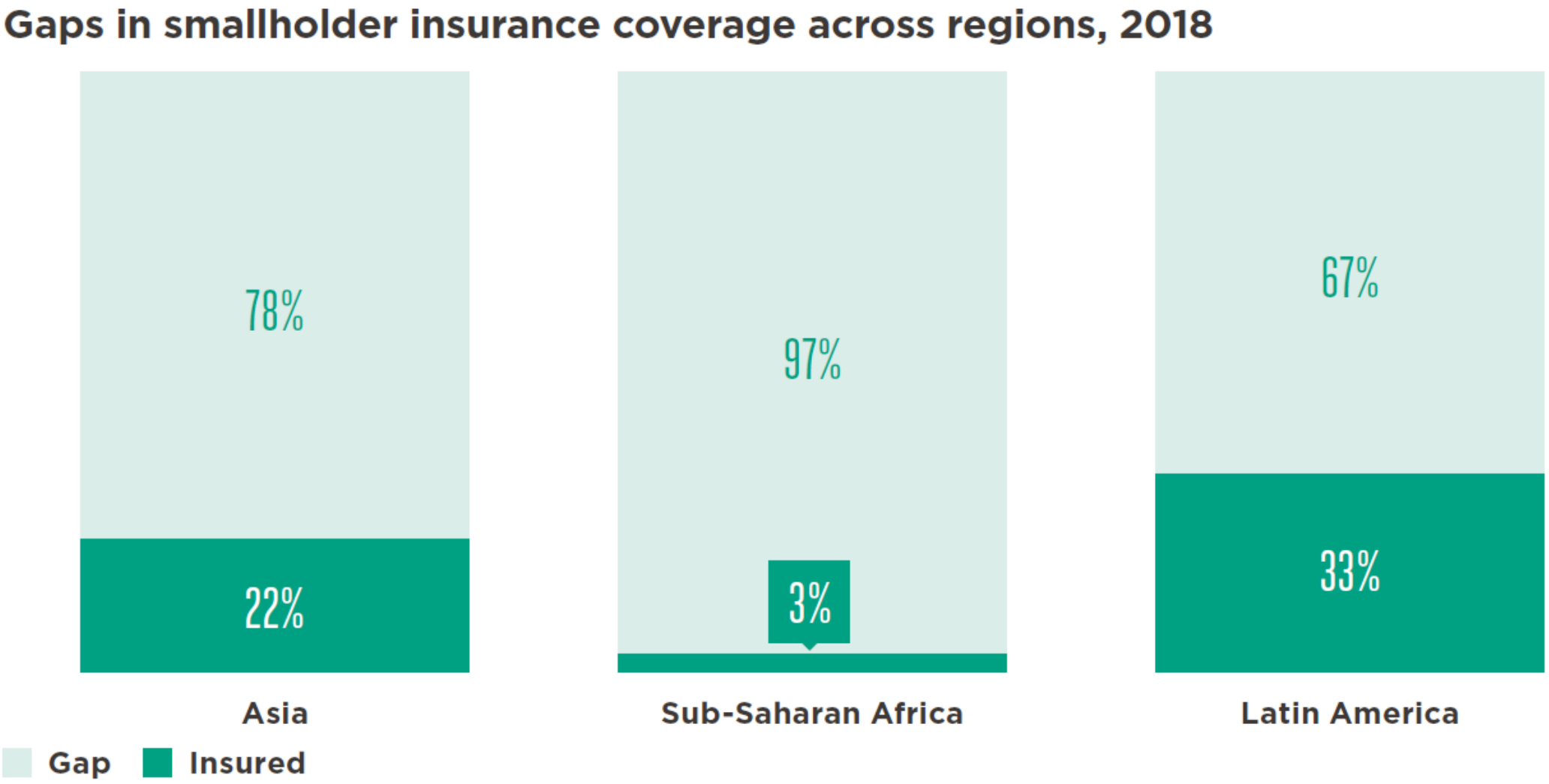
# Gap in smallholder insurance coverage across regions, 2018




Source: ISF Advisors (2018), *Protecting growing prosperity.*

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# Why is agriculture a special line of business?

Successful insurance companies consistently strive for profitable growth, differentiation, and a unique, sustainable market position. Companies struggle for an edge in service or time-to-market.

The nature of the insurance market makes those goals challenging to meet. Below are reasons why agriculture

1. Difficulties in achieving adequate diversification- crops are planted in regions that suffer the same adverse events
2. Asymmetries of information at underwriting- unable to differentiate high and low risk producers. If premiums are set on averages, might attract high risk producers.
3. Geographical dispersion of production making operation and admin costs high
4. Complexity of biological processes of agricultural production which requires skilled and expert. Cause and effect not readily observable.
5. The distribution network is widespread and products are not standard

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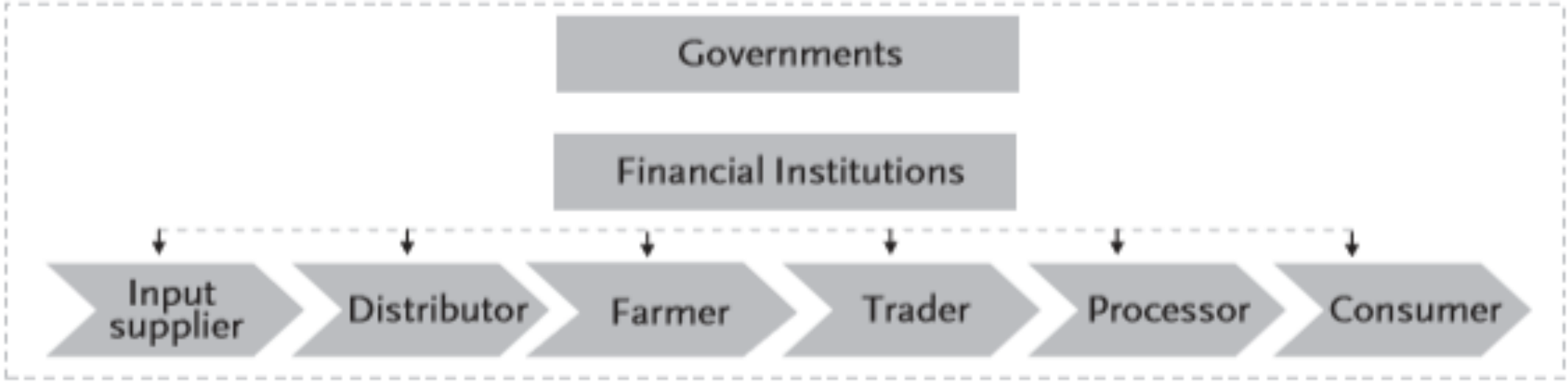
# Agribusiness value chain and risks




| Stakeholder | | Risk |
|---|---|---|
| Goverments | → | Budget Risk/Social Stability |
| Financial Institutions | → | Credit Risk |
| Input Supplier | → | Sales volume / Product enhancement |
| Distributor | → | Sales volume / Product enhancement |
| Farmer | → | Production Risk / Revenue risk |
| Trader | → | Production Risk |
| Processor | → | Lack of Raw Material / Business Interruption |

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# Important financial implications

- The low level of agricultural risk may increase premium rates to unaffordable levels for farmers resulting in risk selection problems
- High loss variance and propensity to catastrophic loss for some perils make premium rates worse
- Premium rates have an important inverse effect on farmer participation
- Loss volatility increases the insurer's dependence on reinsurance. Cooperation with government can help. Example Mongolia set up a risk pooling arrangement for livestock insurance
- **Increased management expenses, increase rates, which decrease sales, which increase adverse selection, which increase loss ratios.**

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# Insurability of agriculture risk

Before an insurer decides to devote capital to agriculture risk, consider the following:

1. Premium charges must be economically feasible for all parties
2. Large number of homogenous subjects exposed to the same risk- assuming individual loss events are independent
3. Probability of the occurrence of insured peril must be calculable
4. Loss should not be catastrophic
5. Loss must be sufficiently large, determinable and measurable- the loss to be insured should be important enough to warrant an insurance contract
6. Loss must be accidental and unintentional

**Challenges:**

1. Because of cyclical nature of farming business, cashflows affect **premium collection**
2. At the set premium, will sufficient economies of scale be achieved to pay claims, increase reserves, pay brokers as well as cater for fixed admin costs?
3. Possible to incur substantial losses even with loss ration below 100%
4. If loss events are not independent- reduce risk retention and possible reduce business to minimize catastrophic risk to portfolio.
5. For catastrophic events- Physical risk reduction, loss minimization and government catastrophe fund or catastrophe insurance pool

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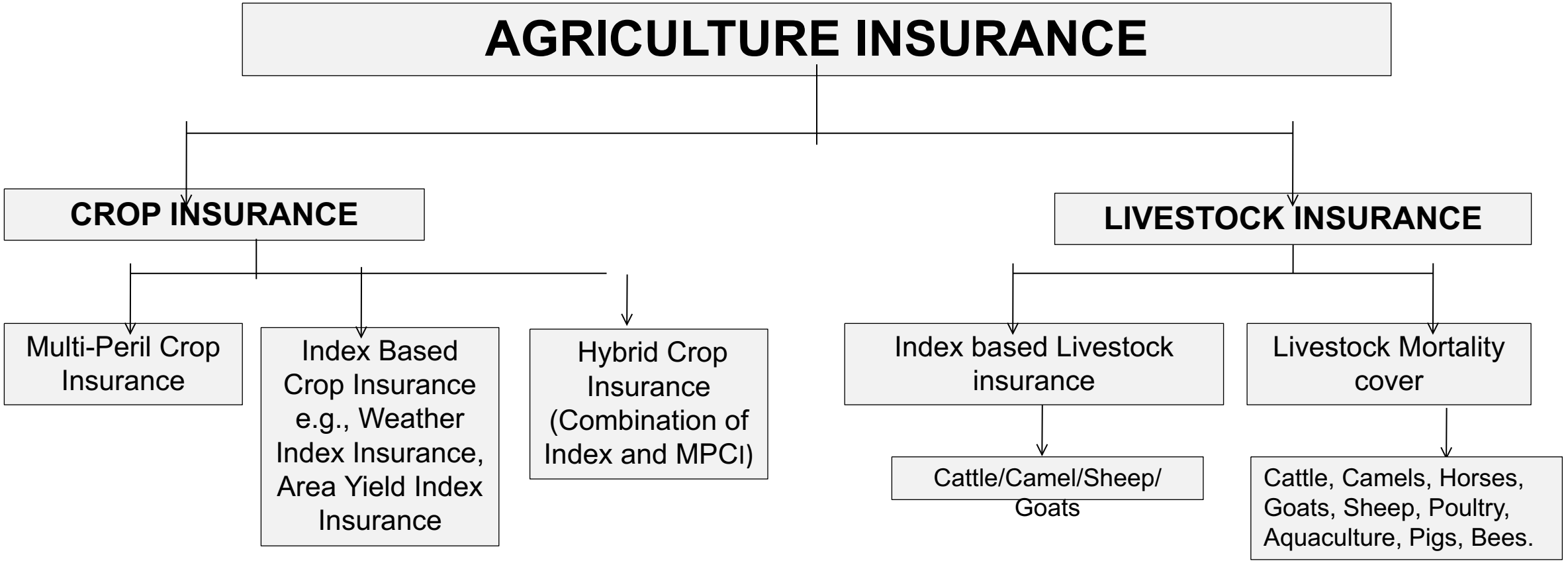
# AGRICULTURE INSURANCE PRODUCTS

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# Importance of Agricultural Insurance

**Risk Mitigation**: Agriculture is susceptible to various risks like weather events, pests, and market fluctuations. Insurance provides a safety net against these uncertainties.

**Income Stabilization**: Insurance payouts compensate for crop losses, stabilizing farmers' income and preventing financial crises during difficult seasons.

**Food Security**: By safeguarding farmers' livelihoods, agriculture insurance contributes to overall food security by ensuring a stable supply of crops.

**Climate Resilience**: In the face of climate change, insurance encourages adoption of climate -resilient practices, promoting long-term sustainability.

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# Importance of Agricultural Insurance cont…..

**Economic Growth**: A robust agriculture sector drives economic growth by boosting production, income generation, and employment opportunities.

**Disaster Recovery**: In the aftermath of disasters, insurance aids in the recovery process, helping communities rebuild faster and more effectively.

**Inclusive Development**: Agriculture insurance extends financial protection to smallholder farmers, fostering inclusivity and reducing inequality in rural areas.

**Debt Management**: Insurance assists farmers in repaying loans even in case of crop failures, reducing the burden of debt and financial strain.

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# What is an Agriculture Insurance Risk

An agriculture insurance risk is any unforeseen event that affects the production of crops and livestock.

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# • Examples of Risks

**Examples of risks**

- **Production risk –** They are natural disasters that occur at farm level. They range from climatic risks to pests and diseases, theft etc.
- **Moral Hazards –** They are the risks that are associated with the money management in the farm, bad financial decisions can be detrimental to the success of a cropping season.
- **Institutional politics risks: -** Risks emanating from government directives.
- **Human or Personnel risk: -** These are risks associated with staff health and personal relationships.
- **Market risks and price volatility:** - These are risks associated with changes in global prices and market forces.

## Examples of Agricultural Insurance risks

- Production risks: - include weather related risks, pests and diseases, theft etc.
- Farm to market risks: - Risks occurring during transportation
- Price volatility

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# Data and Insurance

Data is a critical component for agriculture insurance placement.

The data should be

1. Timely
2. Accurate
3. Relevant
4. Auditable
5. Reliable

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# Types of data sets for Agriculture insurance

Yield data – they are more relatable to farmers, give a more comprehensive coverage but very expensive to collect

2. Weather data or remote sensing data – they offer low cost solutions that have a limited coverage.

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# Underwriting Principles and considerations

Farm management

Diversification

Loss mitigation

Prevention

Insurance

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# Relevance of different categories of crop-insurance

| PRODUCT TYPE | WHEN IS IT POTENTIALLY FEASIBLE? | WHEN IS IT NOT RECOMMENDED? |
|---|---|---|
| 1. Weather Index insurance for Crops | a. Rainfall/weather dependent farming;<br>b. Small-scale farming;<br>c. Good accuracy of underlying weather data | a. Heavily irrigated farming (WII less relevant);<br>b. Accuracy of weather data is not proven;<br>c. Expectation of indemnity loss assessment |
| 2.Area Yield Index Insurance for Crops; | a. Adequate historical granular yield data;<br>b. Yield data is representative of farmers' yield;<br>c. Yield data is available speedily for claims | a. Inadequate historical yield data available;<br>b. Yield data cannot be collected;<br>c. Poor correlation between yield index and farmers' losses<br>d. Concentration risk due to low granularity |
| 3. Indemnity Insurance for Crops | a. Good quality historical yield and loss data is available;<br>b. Loss adjustors or claims management is viable;<br>c. Cost of claims assessment can be absorbed into premium rates;<br>d. Underwriting is possible e.g. of emergence of seedling, farming practices. | a. Typically for smallholder farmers- very difficult to apply without presence of good claims management;<br>b. If historical yield/loss data is not available.<br>c. If farming practices cannot be monitored reliably. |

## • Day 1 Recap

1. Discuss the state of agriculture insurance globally and in Africa
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# • Day 1 Quiz

1. What percentage of gross written premiums does Africa contribute globally
   A. 1%
   B. 10%
   C. 2%
   D. 5%

2. Which of the following is not an example of an agricultural insurable risk
   A. Production Risks
   B. Moral Hazards
   C. Reputation risk
   D. Price Volatility

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# • Quiz

3. Which of the following statement is true

1. A large proportion of small holder farmers are not insured
2. it is not necessary for insurance companies to insure small holder farmers
3. Small holder farmers are cheap to insure
4. We do not have small holder farmers outside Africa

4. Which of the following is not a principle to consider for agricultural insurance underwriting

A. Risk mitigation
B. Diversification
C. Farm management
D. Loss Availability

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# • QUIZ

5. Which of the following statement is true

A. A weather index insurance is appropriate for a non irrigated field
B. An indemnity product is ideal for small holder farmers
C. An Area yield index is applicable if farmers have adequate granular yield
D. Yield is not important in an indemnity product design.

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# Q & A SESSION